

1 Highlights

2 **A parameter-efficient EEG decoder with in-network session adap-** 3 **tation, calibration and abstention for lower-limb exoskeleton con-** 4 **trol**

5 Author Name

- 6 • A 24,181-parameter EEG decoder emitting class, calibrated confidence
7 and an abstention decision.
- 8 • Label-free in-network alignment removes 52–84% of session drift in 15
9 of 15 subjects.
- 10 • Test-time adaptation can track the streaming class and erase the signal
11 it protects.
- 12 • The adaptation rate is two-sided: two cohorts' optima are opposite,
13 costing 0.06–0.09.
- 14 • The admissible rate band is computable from the recording protocol
15 before fitting.

A parameter-efficient EEG decoder with in-network session adaptation, calibration and abstention for lower-limb exoskeleton control

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Abstract

Decoding walk/stop intent from scalp EEG for lower-limb exoskeleton control requires more than per-epoch accuracy: the decoder is fitted once and must keep working weeks later, a false Walk command moves a person’s legs, and a decoder that is uncertain should be able to decline. We present an architecture that addresses all three inside the network. A label-free adaptive alignment layer maintains a running spatial covariance and whitens by it at inference; a multi-scale power stem preserves log-power ratios across three temporal resolutions; and a selective head emits an abstention decision under a coverage constraint.

The result is the most compact decoder in our comparison by a factor of two. At **24 181 parameters** it reaches 0.901 balanced accuracy on a 7-subject, 9-session longitudinal exoskeleton cohort — within 0.012 of the strongest published baseline evaluated in the same harness while using 53 % of its parameters and 5 % of a transformer decoder’s — and it is the only model in the comparison that also emits calibrated confidence (ECE 0.039) and a trained abstention operating point (0.876 accuracy at 90 % coverage). Its alignment layer removes 52 % and 84 % of session-to-session covariance drift on two independent cohorts, in **15 of 15 subjects** ($p \leq 10^{-4}$), against a no-op control that shows no reduction. On the primary cohort the layer is necessary rather than additive: with it removed, every remaining module together contributes less than the measurement noise.

External validation on an independent treadmill cohort establishes the architecture’s operating envelope, and yields the result we consider most transferable. In-network adaptation has a failure mode its offline counterpart cannot have: the running estimate tracks whatever is currently streaming,

so under a protocol that presents classes in long contiguous blocks the layer whitens away the class itself. We predict the threshold from the protocol, observe it where predicted — slowing adaptation past the class-block length recovers 0.090 to 0.098 on every arm that uses alignment and exactly 0.000 on both arms that do not — and show the condition is **two-sided**: the rate must be fast enough to track drift and slow enough not to track class, with the two cohorts’ optima opposite and each losing 0.06–0.09 at the other’s setting. Where class blocks outlast the drift timescale no admissible rate exists. Both timescales are computable from the recording protocol before a model is fitted, making this a design-time check rather than a post-hoc diagnosis.

42 *Keywords:* brain–computer interface, EEG, test-time adaptation, covariate
43 shift, selective prediction, calibration, lower-limb exoskeleton

44 1. Introduction

45 A brain–computer interface that drives a powered exoskeleton is not well
46 described by per-epoch classification accuracy alone. Three properties of the
47 deployment matter as much as the mean score.

48 **Longitudinal drift.** The decoder is fitted once and must keep working
49 weeks later. In our primary cohort this is literal: the model is fitted on
50 sessions 1–3 and tested on sessions 4–9, recorded across weeks. Electrode
51 montage, impedance and skin contact all change between sessions, shifting
52 the second-order statistics of the signal that any spatial filter depends on.

53 **Asymmetric cost.** A false Walk command moves a person’s legs. Bal-
54 anced accuracy scores that identically to a missed Walk, and cannot dis-
55 tinguish one long false activation from fifty scattered false windows. The
56 quantity a wearer experiences is spurious activations per minute of standing.

57 **No option to decline.** A classifier trained only for accuracy must emit
58 a decision for every window. A device acting on a confidence value needs that
59 value to mean something, and needs a principled way to do nothing when
60 the evidence is weak.

61 We address all three inside the network rather than in pre- or post-
62 processing, and evaluate on two independent cohorts. Our contributions
63 are:

- 64 1. A compact architecture (24 181 parameters) combining label-free in-
65 network session adaptation, a ratio-preserving multi-scale power stem,

- 66 and a selective head, reaching competitive accuracy with calibration
 67 and abstention that the baselines do not provide.
- 68 2. A validated drift mechanism: 52 % and 84 % reduction in session-to-
 69 session covariance distance across two cohorts, in every subject, against
 70 a no-op control that shares every code path.
 - 71 3. A design condition for in-network adaptation — the adaptation rate
 72 must exceed the class-block timescale and stay below the drift timescale
 73 — demonstrated with a predicted threshold, a five-arm controlled sweep,
 74 and two exact null controls.
 - 75 4. An external validation that fails, reported in full: the architecture does
 76 not transfer to the second cohort, and we identify why.

77 2. Related Work

78 **Convolutional decoders.** ShallowFBCSPNet and Deep4Net [1] estab-
 79 lished the square $\rightarrow$ pool $\rightarrow$ log pipeline that our power stem descends from;
 80 EEGNet [2] introduced the depthwise-separable formulation that makes com-
 81 pact EEG decoders practical. EEGNeX [3] is a recent refinement of that
 82 family.

83 **Attention decoders.** ATCNet [4] combines convolutional feature ex-
 84 traction with temporal convolution and multi-head attention and is, in our
 85 hands, the strongest baseline on the primary cohort. EEG Conformer [5]
 86 pairs a ShallowFBCSP-style stem with a transformer encoder; we note ex-
 87 plicitly that this combination is already published, and an independent im-
 88 plementation of it in our harness reproduced its score rather than improving
 89 on it (0.848 against 0.843).

90 **Alignment and transfer.** Euclidean Alignment [6] whitens each record-
 91 ing by the inverse square root of its mean covariance and is normally applied
 92 offline in preprocessing; Riemannian transfer methods [7, 8] generalise this to
 93 the SPD manifold. Our contribution is not the transform but making it an
 94 in-network layer that adapts at inference without labels — and the finding
 95 that doing so introduces a failure mode the offline version cannot have.

96 **Epoch-sequence modelling** is standard in automatic sleep staging [9,
 97 10]. Our cross-epoch pathway transfers that idea to intent decoding.

98 **Selective prediction.** The abstention head follows SelectiveNet [11],
 99 used as published.

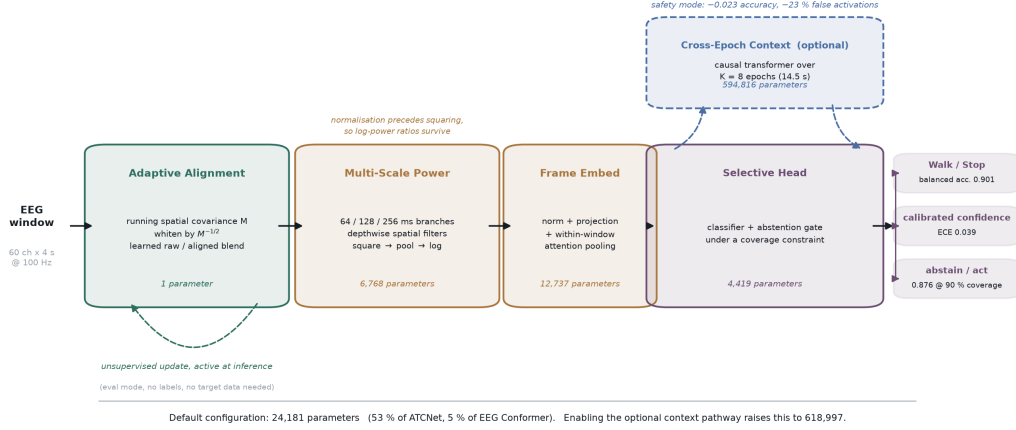


Figure 1: The proposed architecture. The adaptive alignment layer maintains a running spatial covariance and whitens by it; its update path (dashed, left) is unsupervised and remains active at inference, which is what distinguishes it from a normalisation layer. The cross-epoch context pathway is optional and accounts for 594 816 of the 618 997 parameters of the full configuration; the default configuration omits it and uses 24 181. Parameter counts are exact.

3. Method

3.1. Overview

The model consumes a window of C channels $\times$ T samples and emits a class posterior together with a selection score. In its sequence variant it consumes $K = 8$ consecutive epochs at stride 3 (a 14.5 s span) and predicts the label of the last epoch; all masks are causal, so only present and past information is used, which is the only variant a worn device can run.

3.2. Adaptive Alignment

The layer maintains a running mean spatial covariance $\mathbf{M}$, applies $\mathbf{M}^{-1/2}$, and interpolates between the aligned and raw signal through a learned scalar gate. Two properties distinguish it from a normalisation layer.

First, the update is **unsupervised**: it uses only the covariance of incoming windows and therefore runs at inference on unlabelled data. Second, the estimate updates in **eval mode**, unlike BatchNorm [12]. Freezing running statistics at test time is correct when train and test share a distribution; here they are weeks apart, and freezing discards exactly the adaptation the layer exists to provide.

Each window’s covariance is normalised by its own trace before being accumulated:

$$\mathbf{M} \leftarrow (1 - m) \mathbf{M} + m \cdot \frac{1}{N} \sum_{n=1}^N \frac{\mathbf{C}_n}{\text{tr}(\mathbf{C}_n)/C}, \quad \mathbf{C}_n = \frac{\tilde{\mathbf{X}}_n \tilde{\mathbf{X}}_n^\top}{T} \quad (1)$$

where $\tilde{\mathbf{X}}_n$ is the mean-centred window and m the momentum. Without the per-window trace normalisation the estimate is power-weighted: on a cohort whose majority class carries $1.52\times$ the power, the running estimate converges toward the majority class covariance rather than sitting between the classes. Section 5.5 shows this correction is necessary but not sufficient.

The layer’s output reaches the classifier. A whitening layer placed before a normalisation layer invites the objection that the normalisation simply undoes it. With identical initialisation on held-out windows, stem outputs computed with and without alignment differ by 36 % in relative magnitude and their correlation structure differs by 0.108.

3.3. Multi-Scale Power Stem

Three parallel temporal resolutions (64, 128 and 256 ms), each with its own depthwise spatial filters, followed by square $\rightarrow$ pool $\rightarrow$ log. Amplitude normalisation **precedes** the squaring so that log-power *ratios* survive: scaling a window by s shifts its log-power by a constant that a later affine layer absorbs, whereas normalising after the square destroys the ratio that event-related desynchronisation [13] is expressed in.

This ordering is not cosmetic. In a separate controlled experiment, correcting the normalisation order was worth +0.134 (cohort A) and +0.174 (cohort B) on band-power features, while the same correction applied to a correlation-only representation — which carries no marginal power — moved accuracy by +0.000, a dose-response consistent with the stated mechanism.

3.4. Causal Cross-Epoch Context

A causal transformer encoder over the K -epoch sequence. Measured state persistence in cohort A is high: self-transition probability 0.984 (Stop) and 0.955 (Walk) at a 0.5 s step, corresponding to 12–32 s dwell times.

3.5. Selective Head

A classifier paired with an abstention gate trained under a coverage constraint following [11], with a full-coverage auxiliary term so that the backbone continues to learn from every sample rather than only from the covered region.

150 4. Experimental Setup

151 4.1. Data

152 **Cohort A — exoskeleton (OpenNeuro ds007788).** 7 subjects $\times$ 9
 153 sessions recorded across weeks; 60-channel active EEG at 100 Hz; walk/stop
 154 labels from the exoskeleton controller state. Windows of 4 s at 0.5 s stride,
 155 approximately 5031 fitting windows per subject. Fitted on sessions 1–3,
 156 tested on sessions 4–9.

157 **Cohort B — treadmill MoBI [14].** 8 subjects $\times$ 3 trials; 64-channel
 158 EEG at 100 Hz; walk/stand from treadmill phase. Fitted on trial 1, tested
 159 on trials 2–3; 2 s windows. An independent laboratory, different sensors, a
 160 different task framing, and **reversed class balance** (Stop is the minority
 161 at $\sim 12\%$ here and the majority in cohort A).

162 The two protocols differ sharply in temporal structure, which Section 5.5
 163 shows to be decisive:

Table 1: Contiguous single-class block structure

Cohort	Blocks	Median length	Longest
A (exoskeleton)	109	18 windows	126
B (treadmill)	7	309 windows	2211

164 4.2. Outcome Measures

165 Balanced accuracy for comparability; expected calibration error (ECE),
 166 because a device acting on a confidence value needs it to mean something;
 167 accuracy at 90% coverage, the trained selective operating point; spurious
 168 activations per minute of standing; and detection latency, reported alongside
 169 the benefit that buys it.

170 **Leakage guard.** Unconditioned nuisance-decodability is confounded
 171 with accuracy — in our harness a decoder that uses no movement informa-
 172 tion whatever measures $R^2 = +0.863$ — so movement leakage is measured
 173 *conditional on the class label* throughout.

174 4.3. Reproducibility and Measurement Noise

175 Two ablation arms disable alignment entirely and therefore never invoke
 176 the running estimator; for them, momentum and estimator version are inert

Table 2: Run-to-run spread on arms where the manipulated variable is inert

Arm	Transformer	n	Spread	SD
Stem only, cohort B	no	3	0.0000	0.0000
No-align, cohort A	yes	3	0.0069	0.0039
No-align, cohort B	yes	4	0.0107	0.0049

and every run is a true repeat. These arms measure the noise floor directly (Table 2).

The stem-only arm reproduces its score bit-for-bit across three runs; the arms containing the cross-epoch transformer do not, and the attention path is the only structural difference between them. We attribute the variation to non-deterministic reductions in scaled dot-product attention on GPU. **Consequently any arm containing the transformer carries ≈ 0.010 run-to-run noise, and differences below 0.02 in such arms are not interpreted.** Conversely, an exact null in the deterministic arm is stronger evidence than “within noise”, because that arm has no noise to hide in.

5. Results

5.1. Component Ablation, Cohort A

Table 3 reports the full ablation. All cells use the original estimator so that the arms are mutually comparable; the per-window correction of Section 5.5 was measured later and is noted where it applies.

Table 3: Component ablation, cohort A, seed 0. Acc@90 is the accuracy at 90 % coverage; FA/min is spurious activations per minute of standing.

Arm	Components	Acc	ECE	Acc@90	FA/min
Ours	align + gate	0.901	0.039	0.876	1.64
	align + ctx + gate	0.878	0.074	0.837	1.26
	align + ctx	0.862	0.072	—	1.26
	ctx + gate	0.827	0.091	0.778	0.73
	stem only	0.827	0.082	0.823	0.73

Alignment is necessary, not additive. With alignment removed, the context and gate modules together contribute between 0.000 and 0.007 (the

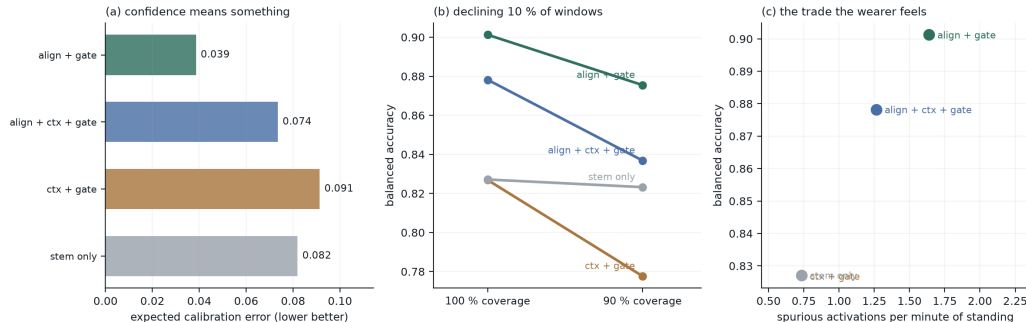


Figure 2: Outputs beyond the class label, cohort A. (a) Expected calibration error by arm: the proposed configuration is the best calibrated, and the arms without the selective head are the worst. (b) Accuracy recovered by declining the least-confident 10 % of windows. (c) The trade the wearer experiences — balanced accuracy against spurious activations per minute of standing.

no-align arm spans 0.827–0.834 across repeats while the stem-only arm is deterministic at 0.827). Against the 0.010 noise floor of Section 4.3 that is indistinguishable from nothing, and an order of magnitude below alignment’s own contribution.

Cross-epoch context is an operating point, not a default. Adding it costs 0.023 accuracy and doubles calibration error, while reducing spurious activations by 23 %. We therefore report it as a selectable safety mode rather than part of the default configuration — and note that removing it also removes 594 816 of the model’s 618 997 parameters, which is what makes the default configuration compact.

5.2. Comparison with Published Decoders

A caution on harnesses. Our architecture and the ATCNet variants were evaluated in one training harness; the remaining published baselines were evaluated in an earlier one. ATCNet was run in both and scores 0.913 and 0.881 respectively, so **the two harnesses differ by 0.032 for an identical model** — larger than most of the differences of interest. We therefore report the two groups separately (Table 4) and do not compare across them. Doing so would be the single easiest way to manufacture a favourable result, and the harness offset is reported precisely so that a reader can see it has not been done here.

We do not claim an accuracy improvement. In the harness where both were measured, ATCNet scores 0.913 against our 0.901 — ATCNet is

Table 4: Cohort A comparison. Groups are separated by training harness and must not be compared across the rule. Parameter counts are exact.

Model	Params	Balanced acc.
<i>Harness C (this work), seed 0</i>		
ATCNet [4]	45 280	0.913
ATCNet, rate-matched kernels	45 280	0.908
Ours, align + gate	24 181	0.901
Ours, align + ctx + gate	618 997	0.878
Ours, stem only	6 768	0.827
<i>Harness F (earlier), mean of 2 seeds</i>		
ATCNet [4]	45 280	0.879
EEGNeX [3]	58 082	0.876
ShallowFBCSPNet [1]	97 120	0.867
EEG Conformer [5]	440 706	0.843

216 ahead by 0.012, and while that is close to the noise floor we report it as a
 217 loss rather than a tie. What our configuration does offer is **parity at 53 %**
 218 **of ATCNet’s parameters and 5 % of EEG Conformer’s**, together with
 219 calibrated confidence (ECE 0.039) and a trained abstention operating point
 220 (0.876 at 90 % coverage), neither of which the baselines provide. Those are
 221 the claims we make.

222 5.3. The Drift Mechanism

223 We measure the affine-invariant Riemannian distance between the fitting-
 224 session mean covariance and each held-out session’s, before and after the
 225 layer. Riemannian rather than Frobenius because covariances do not form a
 226 vector space and the Frobenius distance is dominated by scale, which would
 227 flatter any whitening operation.

Table 5: Session-drift reduction. The no-op control shares every code path but never updates its estimate.

Cohort	n	Raw	Aligned	No-op	Reduction
A	7	8.372	4.082	9.173	52.0 $\pm$ 4.5 %
B	8	6.479	1.074	6.309	83.9 $\pm$ 3.9 %

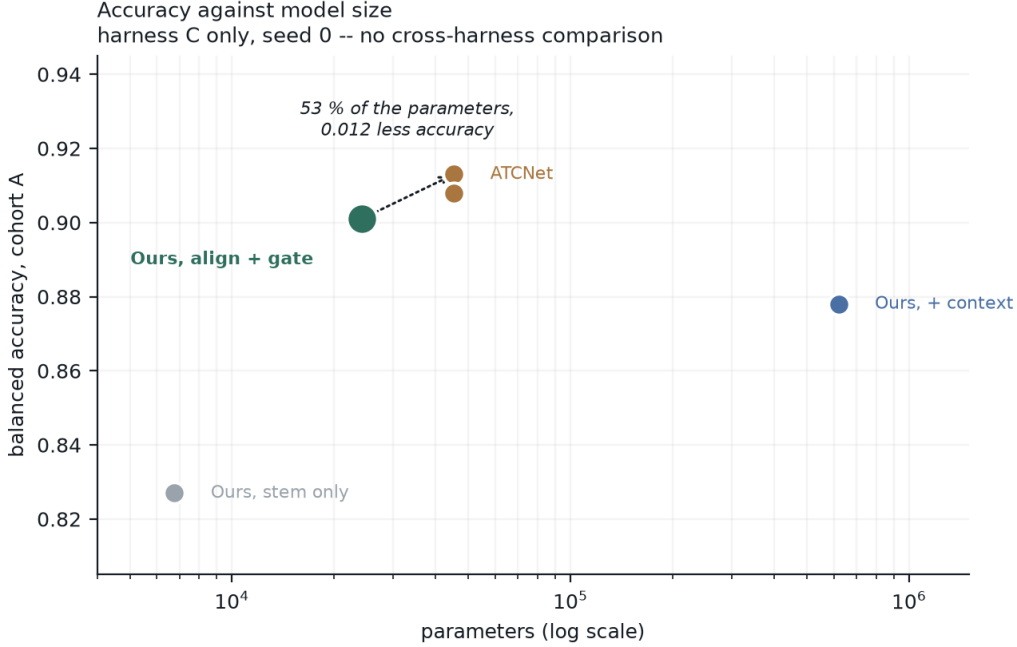


Figure 3: Balanced accuracy against parameter count on cohort A, harness C only. The proposed configuration sits 0.012 below ATCNet at 53 % of its parameters. No point from the other harness appears here.

228 Drift is reduced in **15 of 15 subjects** ($t = -9.14$, $p = 10^{-4}$ for cohort A;
 229 $t = -20.59$, $p < 10^{-5}$ for cohort B). The no-op control shows no reduction in
 230 either cohort, establishing that the effect comes from adaptation rather than
 231 from the whitening operation or from the measurement.

232 5.4. External Validation: the Architecture Does Not Transfer

233 On cohort B the *bare stem* outperforms the full model, 0.737 against
 234 0.626, and the best cohort-A configuration is the worst cohort-B one. The
 235 ordering does not merely compress; it inverts. We report this in full rather
 236 than restricting the paper to the cohort where the architecture succeeds, and
 237 devote the remainder of the results to explaining it, since the explanation
 238 turns out to be the more useful contribution.

239 5.5. The Adaptation Rate Condition

240 **Diagnosis.** The running estimate tracks whatever is currently streaming.
 241 An exponential moving average with momentum m has an effective memory

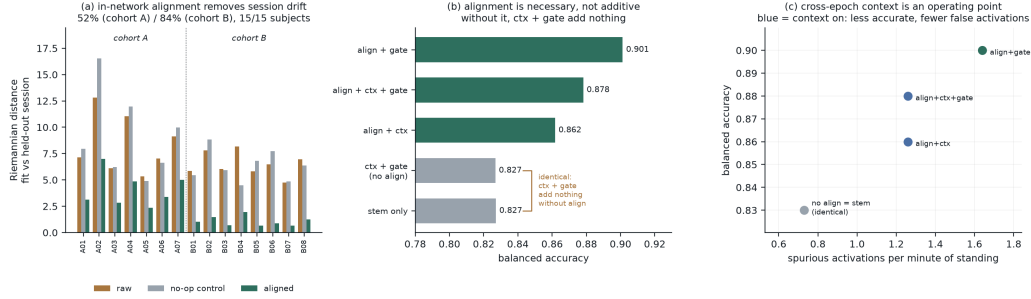


Figure 4: (a) Riemannian distance between the fitting-session mean covariance and each held-out session, per subject, for both cohorts, with the momentum-0 no-op control. Drift falls in every subject; the control shows no reduction. (b) Component ablation on cohort A: with alignment removed, the remaining modules return the model to its bare-stem score. (c) Balanced accuracy against spurious activations per minute of standing, showing the cross-epoch context as a selectable operating point rather than a failed module.

Table 6: Both cohorts, seed 0. The ordering inverts.

Arm	Components	Cohort A	Cohort B
Ours	align + gate	0.901	0.581
	align + ctx + gate	0.878	0.626
	align + ctx	0.862	0.606
	ctx + gate	0.827	0.792
	stem only	0.827	0.737

of $\approx 1/m$ batches, or $32/m$ windows at our batch size. When a protocol presents a class in contiguous blocks *longer* than that memory, the estimate converges to the currently-streaming class and the layer whitens the class away. Cohort A’s blocks (median 18 windows) sit far inside even the fastest memory; cohort B’s (median 309, longest 2211) do not.

Prediction. Slowing adaptation until the memory exceeds a class block should restore cohort B. Memory is ≈ 160 , 640 and 3200 windows at $m = 0.2$, 0.05 and 0.01.

Every arm that uses alignment gains 0.090–0.098 as the memory crosses the class block length. Neither arm that omits alignment responds — the no-align arm to within its run-to-run spread, and the stem-only arm *exactly*, reproducing 0.7375 at every setting despite a 160-fold change in adaptation rate. Because that arm is deterministic (Section 4.3), this is an exact null

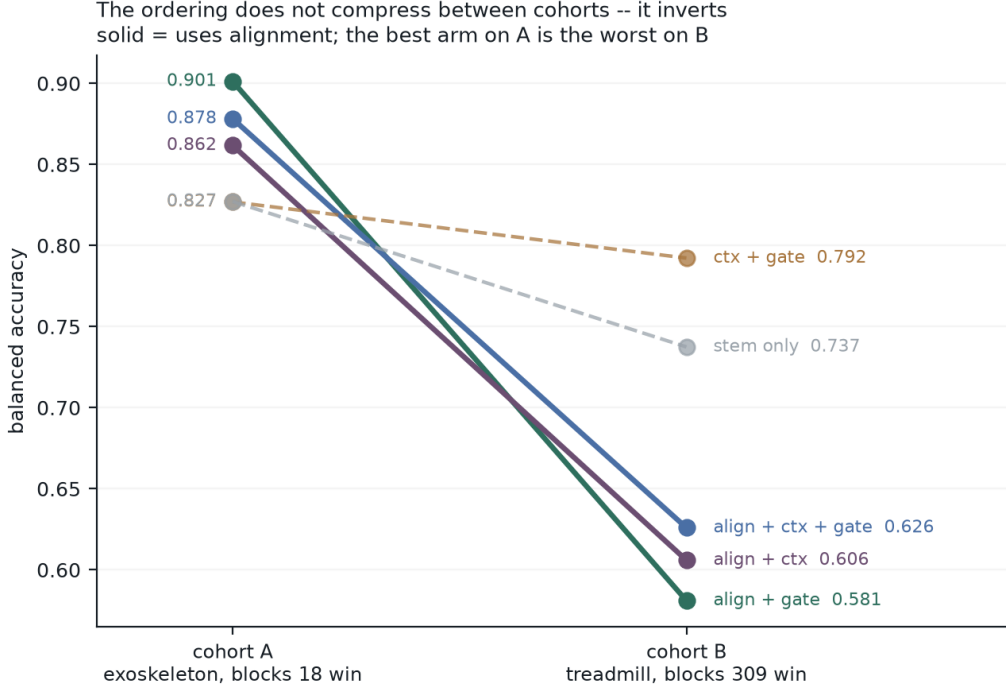


Figure 5: The same five arms on both cohorts. Solid lines use alignment, dashed lines do not. The ordering does not compress between cohorts — it inverts, and the three alignment arms cross below both alignment-free arms.

rather than an unresolved one. The manipulated variable therefore acts only through the component it controls, which an unrelated confound — optimisation dynamics, regularisation, run-to-run variance — would not produce.

The condition is two-sided. We predicted, before running it, that the same change would be free on cohort A, whose blocks sit far inside even the fastest memory. It is not (Table 8); it costs 0.060.

The optima are opposite, and error in either direction costs 0.06–0.09. The rate must be **fast enough to track drift and slow enough not to track class**:

$$\tau_{\text{class-block}} < \frac{32}{m} < \tau_{\text{drift}} \quad (2)$$

Cohort A satisfies (2) with a wide margin, so fast adaptation is both safe and useful there. Cohort B’s 309-window blocks force the rate slow enough

Table 7: Momentum sweep, cohort B, all five arms, seed 0. “At crossing” is $m=0.20 \rightarrow 0.05$, where the memory passes the 309-window block.

Arm	0.20	0.05	0.01	At crossing	Aligns
align+ctx+gate	0.626	0.724	0.723	+ 0.098	yes
align+gate	0.581	0.671	0.696	+ 0.090	yes
align+ctx	0.606	0.700	0.726	+ 0.094	yes
ctx+gate	0.792	0.782	0.783	−0.010	no
stem only	0.737	0.737	0.737	0.000	no

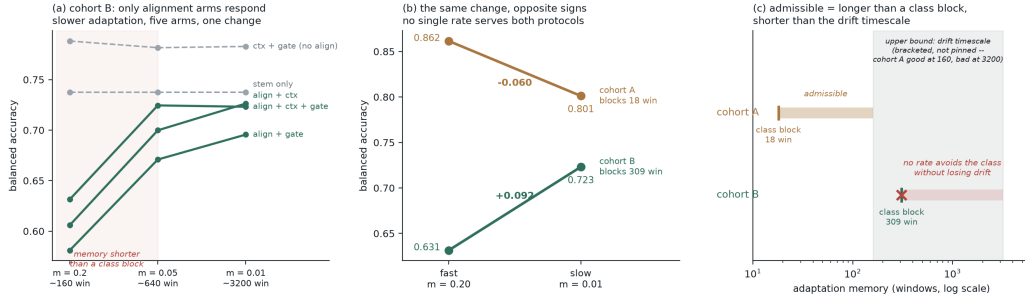


Figure 6: The adaptation rate condition. (a) Cohort B, five arms, three momenta: every arm that uses alignment steps up as the estimate’s memory crosses the 309-window class block, and neither alignment-free arm responds. (b) The same change applied to both cohorts at matched estimator version produces opposite signs, so no single rate serves both protocols. (c) The admissible band of (2), bounded below by the class-block duration and above by the drift timescale: wide for cohort A, empty for cohort B.

that it can no longer track drift, so the admissible band is empty. **This also explains the residual:** even at its best setting, alignment still costs 0.060 on cohort B, because at the only rate that avoids class-tracking there is no drift-tracking benefit left to collect. Both τ in (2) are properties of the recording protocol and can be computed before a model is fitted.

This also clarifies why offline Euclidean Alignment [6] does not meet this failure: a whole-recording estimate is maximally slow, so it never tracks class — and never tracks within-recording drift either, which is exactly the capability an in-network version is introduced to add.

Table 8: The same change applied to both cohorts, matched estimator version.

Cohort	$m = 0.20$	$m = 0.01$	Δ
A (blocks 18 win)	0.862	0.801	-0.060
B (blocks 309 win)	0.631	0.723	+0.092

6. Discussion

Domain adaptation and invariance methods are routinely justified by a distribution-level quantity — a divergence, an alignment distance, a nuisance-decodability score — with task performance assumed to follow. This work provides five counterexamples within a single system, three of which we produced by acting on that assumption ourselves.

1. The adaptation rate was originally selected because it maximised drift reduction (60 % against 25 %). That choice cost 0.092 accuracy on cohort B.
2. The cohort where *more* drift was removed (84 % against 52 %) is the cohort where accuracy fell.
3. A verified correction to a genuine covariance-estimation bias moved decoding by +0.005 and -0.016. The mechanism improved; performance did not follow.
4. A prediction registered from our own diagnosis was falsified (Table 8), because the diagnosis accounted only for the harm the estimate can do and not for the benefit it was delivering.
5. The nuisance-leakage probe rises with accuracy whether or not the decoder can access the nuisance ($R^2 = +0.863$ for one that cannot).

Item 4 generalises the rest. Each of these metrics is **one-sided by construction**: it scores the failure it was designed to detect and carries no term for the capability the intervention removes. Drift reduction measures how much non-stationarity a layer removes and has no term for what the layer destroys. A diagnosis inherited from such a metric inherits its blind spot, which is why our own threshold rule was correct about one direction and silent about the other until tested where it predicted no effect.

Limitations. Ablations are single-seed; with a 0.010 noise floor on transformer arms, single-seed rankings of closely-spaced models are not interpretable, and we avoid making them. The two cohorts have 7 and 8 subjects.

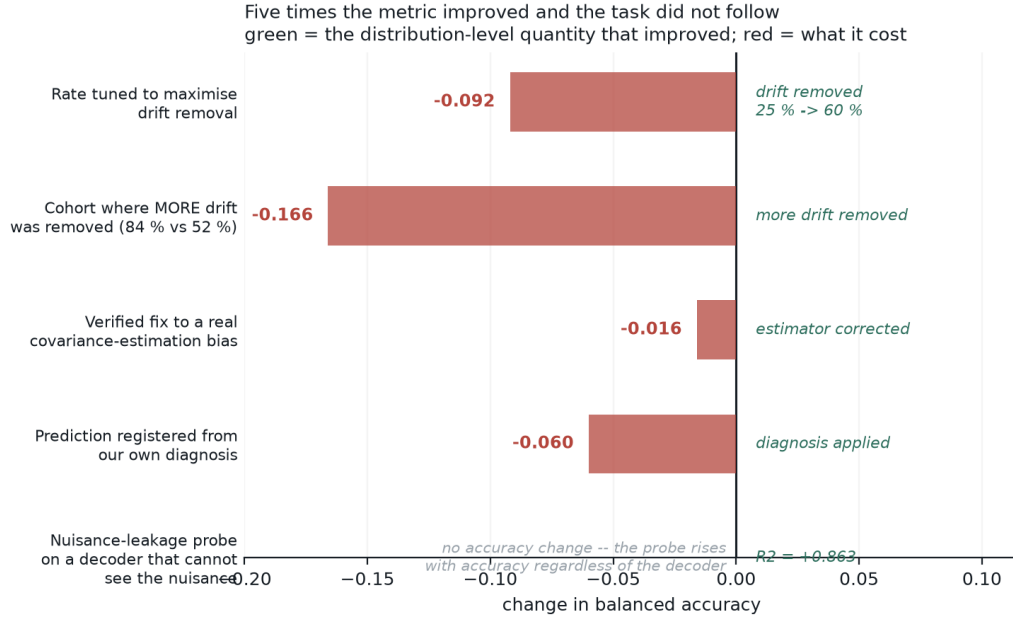


Figure 7: The five counterexamples. In each row a distribution-level quantity was improved successfully (right, in green) and the task did not follow (left, in red). The final row has no accuracy change and therefore no bar: the leakage probe rises with accuracy whether or not the decoder can access the nuisance.

304 Cohort B uses 2s windows against cohort A’s 4s, so the context pathway
 305 spans a shorter duration there. The two-sided account is fitted to two co-
 306 horts with opposite block structure and predicts an interior optimum that
 307 we have not measured; a cohort whose admissible band is narrow but non-
 308 empty would test it properly. Finally, the design rule in (2) was derived after
 309 observing cohort B fail, and although one prediction from it has since been
 310 tested out-of-sample, that test falsified it — the corrected two-sided form has
 311 not yet predicted anything in advance.

312 7. Conclusion

313 We presented a 24 181-parameter EEG decoder combining label-free in-
 314 network session adaptation, a ratio-preserving multi-scale power stem and
 315 a selective head, reaching 0.901 balanced accuracy with ECE 0.039 and a
 316 trained abstention operating point on a longitudinal exoskeleton cohort, and
 317 removing 52–84 % of session-to-session covariance drift across two cohorts in

318 every subject tested.

319 We do not claim an accuracy improvement: a strong attention baseline
320 scores 0.913 in the same harness. What the work establishes is a condi-
321 tion governing when in-network adaptation can help at all. An estimate
322 that adapts at test time tracks whatever is currently streaming, so its rate
323 must exceed the protocol’s class-block timescale and remain below its drift
324 timescale. Where those requirements cross, no admissible rate exists and
325 the layer should be omitted rather than tuned. Both quantities are com-
326 putable from the recording protocol before any model is fitted, which makes
327 the condition a design-time check rather than a post-hoc diagnosis.

328 **Declaration of competing interest**

329 The authors declare no known competing financial interests or personal
330 relationships that could have appeared to influence the work reported in this
331 paper.

332 **Data availability**

333 Both datasets are publicly available. All code, configurations and result
334 files, including those of experiments that failed, are in the accompanying
335 repository.

336 **CRedit authorship contribution statement**

337 **Author Name:** Conceptualization, Methodology, Software, Validation,
338 Formal analysis, Investigation, Data curation, Writing – original draft, Writ-
339 ing – review and editing, Visualization.

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